

3.36pt

Part 5

Inferences about Population Variances

Confidence Interval for Population Variances

Estimation and Tests for Comparing Two Population Variances

INFERENCES ABOUT POPULATION VARIANCES

Estimation and Tests for a Single Population Variance

- Recall that the sample variance $s^2 = \sum (y - \bar{y})^2 / (n - 1)$ is the *point estimate* of σ^2 .
- For tests and confidence intervals about σ^2 we use the fact that the sampling random variable

$$(n - 1) S^2 / \sigma^2 = \chi^2$$

has the *Chi-square Distribution* with $n - 1$ degrees of freedom or *df*, when the sample is from a normal population.

- The chi-square distribution is nonsymmetric.
- Like the Student's t distribution, there is a different curve for each sample size n i.e., for each value of df .
- Percentiles of the chi-square distribution are given in Table 7.
- Plots of the chi-square distribution for $df = 5, 15$, and 30 are shown in Fig. 7.3
- The distribution appear to be more skewed for smaller values of df and become more symmetric as df increases.
- Because the chi-square distribution is nonsymmetric, the percentiles for probabilities at both ends needs to be tabulated.

Chi-square Distribution

FIGURE 7.3
Qualities of the chi-square
 χ^2 = 5, 15, 30 distribution

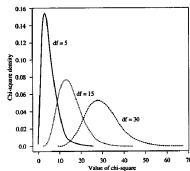


FIGURE 7.4
Critical values of the chi-square distribution with
df = 14

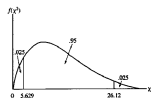
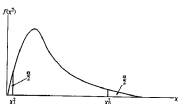


FIGURE 7.5
Upper-tail and lower-tail
values of chi-square



A $100(1-\alpha)\%$ Confidence Interval for σ^2

This has the form:

$$\frac{(n-1)s^2}{\chi_U^2} < \sigma^2 < \frac{(n-1)s^2}{\chi_L^2}$$

where

- Since $df = n - 1$ look up χ_{n-1}^2 percentiles.
- χ_L^2 is the lower-tail value with area $\alpha/2$ to the left.
- χ_U^2 is the upper-tail value with area $\alpha/2$ to the right.
- The confidence interval for the standard deviation σ is found by taking square roots of both end points of above.

Example

Let's assume that the normal probability plot of the data appear to show that the sample is from a normal distribution.

From the data $n = 30$, $\bar{y} = 500.453$, $s = 3.433$ were calculated.

A 99% confidence interval for σ^2 is computed as follows:

Since $\alpha = .1$, $\alpha/2 = .005$ and $1 - \alpha/2 = .995$, we compute

$$\chi_L^2 = \chi_{0.995, 29}^2 = 13.12 \quad \chi_U^2 = \chi_{0.005, 29}^2 = 52.34$$

Thus the endpoints for the required C.I. are, respectively,:

$$\frac{(n-1)s^2}{\chi_U^2} = \frac{29(3.433^2)}{52.34} = 6.53 \quad \frac{(n-1)s^2}{\chi_L^2} = \frac{29(3.433^2)}{13.12} = 26.05$$

Thus (6.53, 26.05) is a 99% C.I. for σ^2

Taking square roots of both endpoints of the interval for σ^2 :

we get (2.56, 5.10) as a 99% C.I. for σ .

Thus we are 99% confident that the standard deviation of the weights of coffee containers lies in the interval (2.56, 5.10).

The filling machine was designed to be so that $\sigma = 4$ grams, and since this value falls in the above interval, the machine satisfies the design specification for standard deviation.

A separate confidence interval for μ is computed for checking whether the design specification for the mean is satisfied.

Tests of Hypotheses about σ^2

Test:

- ① $H_0 : \sigma^2 \leq \sigma_0^2$ vs. $H_a : \sigma^2 > \sigma_0^2$
- ② $H_0 : \sigma^2 \geq \sigma_0^2$ vs. $\sigma^2 < \sigma_0^2$
- ③ $H_0 : \sigma^2 = \sigma_0^2$ vs. $\sigma^2 \neq \sigma_0^2$

Test Statistic: $\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$

Rejection Region: for specified α and $df = n - 1$,

- ① Reject H_0 if $\chi^2 > \chi_U^2$, where $\chi_U^2 \equiv \chi_{\alpha, n-1}^2$
- ② Reject H_0 if $\chi^2 < \chi_L^2$, where $\chi_L^2 \equiv \chi_{1-\alpha, n-1}^2$
- ③ Reject H_0 if $\chi^2 > \chi_U^2$, where $\chi_U^2 \equiv \chi_{\alpha/2, n-1}^2$, or if $\chi^2 < \chi_L^2$, where $\chi_L^2 \equiv \chi_{1-\alpha/2, n-1}^2$

Handout - Example 7.2

Notes about Example

- The above inferences are based on the assumption of sampling from a normal population.
- They are more sensitive to departures from normality than the inferences about population mean.
- We don't have a CLT-type theorem which applies to S^2 like we have in case of $\bar{Y}$.
- Always plotting sample data as a preliminary procedure - using boxplot or normal probability plot to look for skewness, or outliers, is recommended.

Estimation and Tests for Comparing Two Population Variances

- We will consider the case of independent samples from two populations having variances σ_1^2 and σ_2^2 .
- The main application is testing $\sigma_1^2 = \sigma_2^2$.
- Theory says that when the two populations are Normal, and sample sizes are n_1 and n_2 then the random variable

$$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}$$

has the F distribution with degrees of freedom $df_1 = n_1 - 1$ and $df_2 = n_2 - 1$.

Estimation and Tests for Comparing Two Population Variances - cont.

- df_1 is the degrees of freedom associated with S_1^2 .
- df_2 is the degrees of freedom associated with S_2^2 .
- The F distribution is a two parameter family indexed by df_1 and df_2 .
- Some use the terms **Numerator** and **Denominator** degrees of freedom for df_1 and df_2 .
- The F-statistic is calculated as

$$F_c = \frac{s_1^2/\sigma_1^2}{s_2^2/\sigma_2^2}$$

- The F table gives percentiles of the F-distribution only for upper tail areas.

Estimation and Tests for Comparing Two Population Variances - cont.

- The lower tail values, when needed can be obtained by the relationship

$$F_{1-\alpha, df_1, df_2} = \frac{1}{F_{\alpha, df_1^*, df_2^*}}$$

where $df_1^* = df_2$ and $df_2^* = df_1$.

- That is, to calculate a left tail probability, look-up the corresponding right tail probability after switching the numerator and denominator degrees of freedom.
- For example $F_{.95, 3, 10}$ is calculated using the above relationship as $1/F_{.05, 10, 3} = 1/8.79 = 0.11$
- And $F_{.95, 10, 3}$ is calculated using the above relationship as $1/F_{.05, 3, 10} = 1/3.71 = 0.27$

Tests comparing two variances

Test:

1. $H_0 : \sigma_1^2 \leq \sigma_2^2$ vs. $H_a : \sigma_1^2 > \sigma_2^2$
2. $H_0 : \sigma_1^2 = \sigma_2^2$ vs. $H_a : \sigma_1^2 \neq \sigma_2^2$

Test Statistic:

$$F = \frac{s_1^2}{s_2^2}$$

Rejection Region: For given α , $df_1 = n_1 - 1$, and $df_2 = n_2 - 1$.

1. Reject H_0 if $F \geq F_{\alpha, df_1, df_2}$
2. Reject H_0 if $F \leq F_{1-\alpha/2, df_1, df_2}$ or $F \geq F_{\alpha/2, df_1, df_2}$

Example

In order to test the hypothesis $H_0 : \mu_1 - \mu_2 = 0$ using a two-sample t -statistic with pooled sample variance, we need to assume $\sigma_1^2 = \sigma_2^2$.

To check whether this is a valid assumption, let us formally test

$$H_0 : \sigma_1^2 = \sigma_2^2 \text{ vs. } H_a : \sigma_1^2 \neq \sigma_2^2$$

using $\alpha = .05$

The two independent samples gave sample variances:

$$s_1^2 = 0.105, \quad s_2^2 = 0.058 \text{ with } df_1 = 9 \text{ and } df_2 = 9.$$

Example - cont.

Test: $H_0 : \sigma_1^2 = \sigma_2^2$ vs. $H_a : \sigma_1^2 \neq \sigma_2^2$

Test Statistic: $F_c = \frac{s_1^2}{s_2^2} = \frac{0.105}{0.058} = 1.81$

Rejection Region: $F \leq F_{.975,9,9}$ or $F \geq F_{.025,9,9}$

From Table 8, using $\alpha = 0.05$, $df_1 = 9$, $df_2 = 9$, we have

$F_{0.025,9,9} = 4.03$ and $F_{0.975,9,9} = 1/F_{0.025,9,9} = 1/4.03 = 0.25$

Thus we reject if $F \leq 0.25$ or $F \geq 4.03$.

Since $F_c = 1.81$ does not fall in the rejection region, we fail to reject $H_0 : \sigma_1^2 = \sigma_2^2$.

Conclusion: The assumption of equal variances for the two populations appears reasonable.

Test comparing several variances

Hartley's F_{max} Test for Homogeneity of Variances

This test requires that independent random samples of equal sample sizes, n , are drawn from t populations having normal distributions.

Test: $H_0 : \sigma_1^2 = \sigma_2^2 = \dots = \sigma_t^2$ vs. H_a : Not all σ^2 's equal

Test Statistic: $F_{max} = \frac{s_{max}^2}{s_{min}^2}$ where s_{max}^2 = largest sample variance and s_{min}^2 = smallest sample variance

Rejection Region: For given α , reject H_0 if F_{max} exceeds the tabulated percentile from Table 12, for α .

Note: Use t , $df_2 = n - 1$ and α to read Table 12

Test comparing several variances cont.

- When **sample sizes are not all equal** an approximate test is obtained if n is replaced with n_{max} , where n_{max} is largest sample size, in the above procedure.
- Hartley's test is quite sensitive to violation of the normality assumption.
- Thus other procedures that do not require the normality assumption, such as Levene's test has been proposed.
- Levene's test is too cumbersome to calculate by hand; so software may have to be used.
- Levine's test procedure is less powerful than Hartley's test when populations have normal distributions.